1. Capillary theory

Boehm (1809) suggested that the xylem vessels work like a capillary tube. This capillarity of the vessels under normal atmospheric pressure is responsible for the ascent of sap. This theory was rejected because the magnitude of capillary force can raise water level only up to a certain height. Further, the xylem vessels are broader than the tracheid which actually conducts more water and against the capillary theory.

2. Imbibition theory

This theory was first proposed by **Unger** (1876) and supported by **Sachs** (1878). This theory illustrates, that water is imbibed through the cell wall materials and not by the lumen. This theory was rejected based on the ringing experiment, which proved that water moves through the lumen of the cell and not by a cell wall.

3. Cohesion-tension or Cohesion and transpiration pull theory

Cohesion-tension theory was originally proposed by **Dixon** and **Jolly** (1894) and again put forward by **Dixon** (1914, 1924). This theory is based on the following features:

i. **Strong cohesive force or tensile strength of water**

Water molecules have the strong mutual force of attraction called **cohesive force** due to which they cannot be easily separated from one another. Further, the attraction between a water molecule and the wall of the xylem element is called **adhesion**. These cohesive and adhesive force works together to form an unbroken continuous water column in the xylem. The magnitude of the cohesive force is much high (350 atm) and is more than enough to ascent sap in the tallest trees.

ii. **Continuity of the water column in the plant**

An important factor which can break the water column is the introduction of air bubbles in the xylem. Gas bubbles expanding and displacing water within the xylem element is called **cavitation** or **embolism**. However, the overall continuity of the water column remains undisturbed since water diffuses into the adjacent xylem elements for continuing ascent of sap.

iii. **Transpiration pull or Tension in the unbroken water column**

The unbroken water column from leaf to root is just like a rope. If the rope is pulled from the top, the entire rope will move upward. In plants, such a pull is generated by the process of transpiration which is known as **transpiration pull.**

Water vapour evaporates from mesophyll cells to the intercellular spaces near stomata as a result of active transpiration. The water vapours are then transpired through the stomatal pores. Loss of water from mesophyll cells causes a decrease in water potential. So, water moves as a pull from cell to cell along the water potential gradient. This tension, generated at the top (leaf) of the unbroken water column, is transmitted downwards from petiole, stem and finally reaches the roots. The cohesion theory is the most accepted among the plant physiologists today.

11.6 Transpiration

Water absorbed by roots ultimately reaches the leaf and gets released into the atmosphere in the form of vapour. Only a small fraction of water (less than 5%) is utilized in plant development and metabolic process.

The loss of excess of water in the form of vapour from various aerial parts of the plant is called **transpiration**. Transpiration is a kind of evaporation but differs by the involvement of biological system. The amount of water transpired is astounding (Table 11.4). The water may move through the xylem at a rate as fast as 75cm /min.

Table: 11.4 Rate of Transpiration in some plants

Plant	Transpiration per day
Corn plant	2 Litres
Sunflower	5 Litres
Maple tree	200 Litres
Date palm	450 Litres

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